STUDENTS' TECHNICAL COUNCIL

AccioJob

14-16 APRIL 2023

TECHSHILA

INTER BHAWAN TECH MEET

Natural Language Processing

#3

AccioJob's Multilingual Video Translation Challenge

Are you ready for a challenge that will test your AI-pipeline skills and make a difference in education? Join the Hinglish to Telugu dubbing challenge and help learners access programming courses in their native language. You will be creating a dubbed video that sounds natural and engaging in Telugu, using the latest technologies and datasets. You will also be learning from the best in the field and competing with other talented teams. This is your chance to solve a real-world problem, showcase your creativity, and have fun!

Problem Statement

Imagine you are a software engineer who wants to learn Python programming. You have found a great video course on YouTube that teaches Python in a fun and interactive way. But there is a catch: the instructor speaks in “Hinglish”, a blend of Hindi and English, which you are not familiar with. You wish you could watch the same video with the instructor’s voice in your native language, Telugu. That’s where you come in. You have heard about the Hinglish to Telugu dubbing challenge, and you want to help. Your goal is to create an AI-pipeline that can take any video with an instructor teaching programming in “Hinglish” and turn it into a dubbed video in “Telugu” that sounds natural and engaging. By doing this, you will be able to help learners like yourself to access high-quality educational content in their preferred language.

Task

Your task is to design, train, and deploy an AI-pipeline that can seamlessly translate and dub the speech of an instructor teaching programming in “Hinglish” into “Telugu” while preserving the voice quality, emotion, and context of the original video. The pipeline should consist of three main components:

- A speech recognition model that can transcribe the speech from the input video into text.
- A machine translation model that can translate the text from “Hinglish” to “Telugu”.
- A speech synthesis model that can generate speech from the translated text in “Telugu” with the same voice characteristics as the instructor.

The pipeline should be able to process a video within 5 times its duration on a 4vCPU 8GB RAM machine.

For example, a 2-hour lecture should be dubbed in no more than 10 hours on the given machine configuration. The output video from the pipeline should contain a human-like speech in “Telugu” that matches the pitch, tone, naturalness, and intonation of the instructor in the original video. The output video should also be synchronized with the lip movements and gestures of the instructor to avoid any mismatch or confusion.

Evaluation

The quality of the output video will be evaluated based on two criteria: voice quality (70% weightage) and WER (30% weightage).

- Voice quality refers to how well the output speech sounds like the instructor's voice and how natural and expressive it is.
- WER (Word Error Rate) refers to how accurately the output speech matches the meaning and intent of the original speech.

The points distribution is as follows:

1. Total Event Points: 100
2. Voice Quality weightage: 70%
3. WER weightage: 30%

Participation

Participation details related to the challenge:

1. The maximum team size is 6 members.
2. Contact your bhawan's contingent leader to participate.

Submission

You need to submit a well-formatted and well-annotated iPython Notebook which should clearly explain your

approach, your model, etc. Alongside, you will also be required to submit a demo working of the application, clearly showing the translation from Hinglish to Telugu. Details regarding the submission will be communicated via discord once the teams have been finalized. The submission details are as follows:

- 13th April: Code+Demo Submission
- 14th April: Presentation

Resources and Conclusion

Hinglish to Telugu dubbing is an exciting field of research that has the potential to democratize education and make it accessible to everyone. With the help of AI-pipeline, you can create dubbed videos that sound natural and engaging in any language. Here are some resources to help you out:

- OPEN AI Whisper: A state-of-the-art speech recognition model that can transcribe speech from any language or accent. (DEMO) (paper) (implementation)
- Microsoft VALL-E: A human-like speech synthesis model that can generate speech from text in multiple languages and voices. (DEMO) (paper) (unofficial implementation)
- INDIC TTS: IIT Madras's text-to-speech implementation implementation in Indian languages with datasets (link)
- Indian Languages text to audio datasets, Demos and pre-trained models (link)